

非精确线搜索

Armijo 准则: $f(x^k + \alpha \phi^k) \leq f(x^k) + c_1 \alpha \langle \nabla f(x^k), \phi^k \rangle$
梯度下降 $f(x^k - \alpha \nabla f(x^k)) \leq f(x^k) - c_1 \alpha \|\nabla f(x^k)\|^2$

Wolfe 准则:
$$\begin{cases} f(x^k + \alpha \phi^k) \leq f(x^k) + c_1 \alpha \langle \nabla f(x^k), \phi^k \rangle \\ \langle \nabla f(x^k + \alpha \phi^k), \phi^k \rangle \geq c_2 \langle \nabla f(x^k), \phi^k \rangle \end{cases}$$

计算复杂度:

Flop 计数: 两个浮点数之间的一次四则运算.

① 向量求和 $x, y \in \mathbb{R}^n$ $x+y \rightarrow n \text{ flops}$

② 向量内积 $x, y \in \mathbb{R}^n$ $x^T y \rightarrow 2n-1 \text{ flops}$

③ 标量乘法 $\alpha \in \mathbb{R} \ x \in \mathbb{R}^n$ $\alpha x \rightarrow n \text{ flops}$

④ 矩阵乘法 $y = Ax$ $A \in \mathbb{R}^{m \times n}$ $x \in \mathbb{R}^n \rightarrow m(2n-1) \text{ flops}$

⑤ 矩阵相乘 AB $A \in \mathbb{R}^{m \times n}$ $B \in \mathbb{R}^{n \times p}$
 $A \begin{bmatrix} | & | & & | \\ b_1 & b_2 & \dots & b_p \\ | & | & & | \end{bmatrix} \rightarrow m(2n-1)p \text{ flops}$
 $\approx 2mnp$

B.x. $A \in \mathbb{R}^{1000 \times 100}$ $B \in \mathbb{R}^{100 \times 2000}$ $C \in \mathbb{R}^{2000 \times 100}$
 $D = A \cdot B \cdot C$ 8×10^8

$A \cdot B$ $2 \times 1000 \times 100 \times 2000 \rightarrow E \in \mathbb{R}^{1000 \times 2000}$
 4×10^8

$E \cdot C$ $2 \times 1000 \times 2000 \times 100$ 4×10^8

$4 \times 10^8 + 4 \times 10^8 \approx 8 \times 10^8$

$$D = A \cdot (B \cdot C)$$

$$E = B \cdot C : 2 \times 100 \times 2000 \times 10 = 4 \times 10^6 \text{ FLOPs}$$

$$A \cdot E \approx 2 \times 1000 \times 100 \times 10 = 2 \times 10^6$$

6×10^6

⑤.① 对称矩阵相乘 $C = A \cdot A^T$ $A \in \mathbb{R}^{m \times n}$

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & \dots & c_{1n} \\ c_{12} & \ddots & \ddots & \ddots & \ddots \\ \vdots & \ddots & \ddots & \ddots & \ddots \\ c_{1n} & \ddots & \ddots & \ddots & c_{nn} \end{bmatrix}$$

→ 共有 $\frac{1}{2}(1+m)m$ 个元素要计算,

$c_{ij} = A_i^T A_j$: A 中第 i 行与第 j 行相乘

$2n-1$ flops,

∴ 共有 $\frac{1}{2}(1+m)m(2n-1)$ flops

$\approx m^2 n$ flops.

⑤.② 稀疏矩阵: A 有 N 个非零元素, $\rightarrow 2N$ flops

$$A = \begin{pmatrix} 3 & 0 & 0 & 5 \\ 0 & 0 & -2 & 0 \\ 0 & 7 & 0 & 0 \end{pmatrix} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

$$y = Ax \quad y = \begin{pmatrix} 0 + 7x_1 + 5x_4 \\ 0 + (-2)x_3 \\ 0 + 7x_2 \end{pmatrix}$$

⑤.③ 低秩矩阵: 已知 $A = UV^T$ $A \in \mathbb{R}^{m \times n}$ $U \in \mathbb{R}^{m \times p}$ $V \in \mathbb{R}^{n \times p}$

$$y = Ax = UV^T x = U(V^T x)$$

$m \gg p, n \gg p$

$$z = V^T x \quad p(2n-1) \approx 2pn \text{ flops}$$

$$y = Uz \quad m(2p-1) \approx 2pm \text{ flops}$$

$$\rightarrow 2p(n+m) \text{ flops}$$

$m(2n-1)$ flops

$\approx 2mn$ flops

梯度下降法的计算量

$$f(x) = \frac{1}{2} x^T A x - b^T x \quad x \in \mathbb{R}^n$$

$$x^{k+1} = x^k - \alpha \nabla f(x^k)$$

每一步迭代的计算量

$$2n(n+1) \text{ flops} \left\{ \begin{array}{l} \nabla f(x^k) = Ax - b \rightarrow n(n-1) + n = 2n^2 \text{ flops} \\ x^{k+1} = x^k - \alpha \nabla f(x^k) \rightarrow 2n \text{ flops} \end{array} \right.$$

总计算量: $2kn^2$

对于 $f(x) = \frac{1}{2}x^T Ax - b^T x$

$$\nabla f(x) = 0 \Rightarrow Ax - b = 0 \Rightarrow x = A^{-1}b$$

求解 $Ax = b$ 的计算量

① 最简单的方法 $x = A^{-1}b$

设 A 为非奇异矩阵

(i) 求 A^{-1} : Gaussian - Jordan 消元法

$$[A|I] \Rightarrow [I|A^{-1}]$$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \quad \begin{array}{l} \text{① 把 } a_{11} = 1 \\ \text{2n个除法} \\ \text{② 把第k列其它元素} \\ \text{化为0} \\ \text{n-1行, 每行2n个加法} \\ \text{2n(n-1) + n} \end{array}$$

$$\text{总计算量: } n(2n(n-1) + n + 2n) \approx 2n^3 \text{ flop}$$

$$2n(n-1)$$

$$\|Ax - b\|^2 = 0$$

$$\Leftrightarrow Ax = b$$

$$\min_x \|Ax - b\|^2$$

$$f(x) = x^T A^T A x - 2b^T A x + b^T b$$

$$\Rightarrow \nabla f(x) = 2A^T A x - 2b^T A = 0$$

$$\Rightarrow x = (A^T A)^{-1} A^T b$$

$$A \in \mathbb{R}^{m \times n} \quad m \geq n$$

A 列满秩

LU分解

② 带状结构的矩阵A

(1) 对角矩阵

$$x = A^{-1}b$$

$$A = \begin{bmatrix} a_{11} & & \\ & \ddots & \\ & & a_{nn} \end{bmatrix} \quad A^{-1} = \begin{bmatrix} \frac{1}{a_{11}} & & \\ & \ddots & \\ & & \frac{1}{a_{nn}} \end{bmatrix} \quad n \text{ 除法} + n(2n-1) \rightarrow 2n^2 \text{ flops}$$

(2) 下三角矩阵

$$x_1 = b_1 / a_{11}$$

1 flop

$$A = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$x_2 = (b_2 - a_{21}x_1) / a_{22} \quad 3 \text{ flops}$$

$$x_3 = (b_3 - a_{31}x_1 - a_{32}x_2) / a_{33} \quad 5 \text{ flops}$$

$$x_k = (b_k - a_{k1}x_1 - \dots - a_{kk-1}x_{k-1}) / a_{kk} \quad 2k-1 \text{ flops}$$

一共 $\frac{1}{2}(1+2n-1)n = n^2$ flops 计算量。

(3) 上三角矩阵：同上 n^2 flops.

(4) 正交矩阵： $AA^T = I \Rightarrow A^{-1}AA^T = A^{-1} \Rightarrow A^T = A^{-1}$

$$Ax = b \Rightarrow x = A^{-1}b = A^T b \rightarrow n(2n-1) \text{ flops}$$

③ LU分解求解 $Ax = b$

$$\text{LU分解} \quad A = LU$$

步骤 (i) A分解为 $A = LU$

$$\rightarrow \frac{2}{3}n^3 \text{ flops}$$

(ii) 求解 $Lz = b$

$$\rightarrow n^2 \text{ flops}$$

(iii) 求解 $Ux = z$

$$\rightarrow n^2 \text{ flops}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

(i) 算乘子

$$l_{21} = \frac{c_{21}}{a_{11}} \quad l_{31} = \frac{a_{31}}{a_{11}}$$

$n-1$ 次除法

$$l_{32} = \frac{a_{32}}{a_{22}}$$

$n-2$ 次除法

!

$n-k$ 次除法

$n-1$ ↑

(ii) 消元

$$(a_{21} \ a_{22} \ a_{23}) - l_{21} (a_{11} \ a_{12} \ a_{13}) \rightarrow (0 \ u_{22} \ u_{23})$$

→ $n-1$ 次乘法, $n-1$ 次减法

$$(a_{31} \ a_{32} \ a_{33}) - l_{31} (a_{11} \ a_{12} \ a_{13}) \rightarrow 2n-2$$

$$= (0 \ a'_{32} \ a'_{33})$$

$$(0 \ a'_{32} \ a'_{33}) - l_{32} (a_{21} \ a_{22} \ a_{23}) \rightarrow 2n-4$$

$$= (0 \ 0 \ u_{33})$$

$$\text{总计算量: } \sum_{k=1}^{n-1} [(n-k) + 2(n-k)^2]$$

$$\text{令 } r = n-k \Rightarrow \sum_{r=1}^{n-1} (r + 2r^2)$$

$$\sum_{r=1}^{n-1} r = \frac{1}{2} n(n-1)$$

$$\sum_{r=1}^{n-1} r^2 = \frac{1}{6} (n-1)n(2n-1)$$

$$\Rightarrow \frac{2}{3} n^3 \text{ flops}$$

复杂度: $2n^3 \text{ flops}$

④ cholesky 分解
 $A = LL^T$

设 A 为正定矩阵

步骤 (i) cholesky 分解

$\rightarrow \frac{1}{2}n^2$ flops

(ii) 求 $Lz_1 = b$

$\rightarrow n^2$ flops

(iii) 求 $L^T x = z_1$

$\rightarrow n^2$ flops

梯度下降 $O(n^2)$

共轭梯度法：用求解 $Ax = b$ ，要求 A 为正定矩阵

Defn (共轭)：若向量 u, v 关于矩阵 A 共轭，那么
 $\langle u, Av \rangle = 0$

Intuition：搜索当前梯度方向的共轭方向进行迭代

算法：共轭梯度法

初始化： x^0 初始点， $z \in (0, 1)$

1. $r^0 = b - Ax^0$

2. $d^0 = r^0$

3. while $\|r^{k+1}\| > \epsilon$

4. 步长 $\alpha_k = (r^k)^T r^k / (d^k)^T d^k$

5. 迭代点 $x^{k+1} = x^k + \alpha_k d^k$

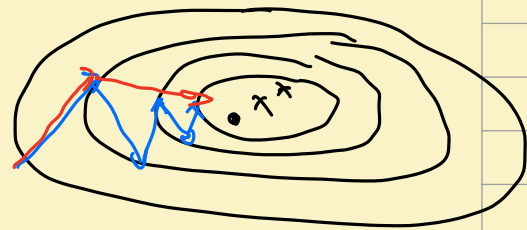
6. $r^{k+1} = r^k - \alpha_k d^k$

7. $\beta_k = (r^{k+1})^T r^{k+1} / (r^k)^T r^k$

8. $d^{k+1} = r^{k+1} + \beta_k d^k$

9. End while

— 梯度下降法
 — 共轭梯度法



计算量: $\sqrt{\kappa(A)} \sim^2 \text{ flops.}$

Thm: 对于严格凸的二次函数 $f(x) = \frac{1}{2} x^T A x - b^T x$, 共轭梯度法产生的迭代序列满足 → 迭代次数

$$\|x^k - x^*\|_A \leq \left(\frac{\kappa(A) - 1}{\kappa(A) + 1} \right)^k \|x^0 - x^*\|_A$$

梯度下降法:

$$\|x^k - x^*\|_A \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|x^0 - x^*\|_A$$